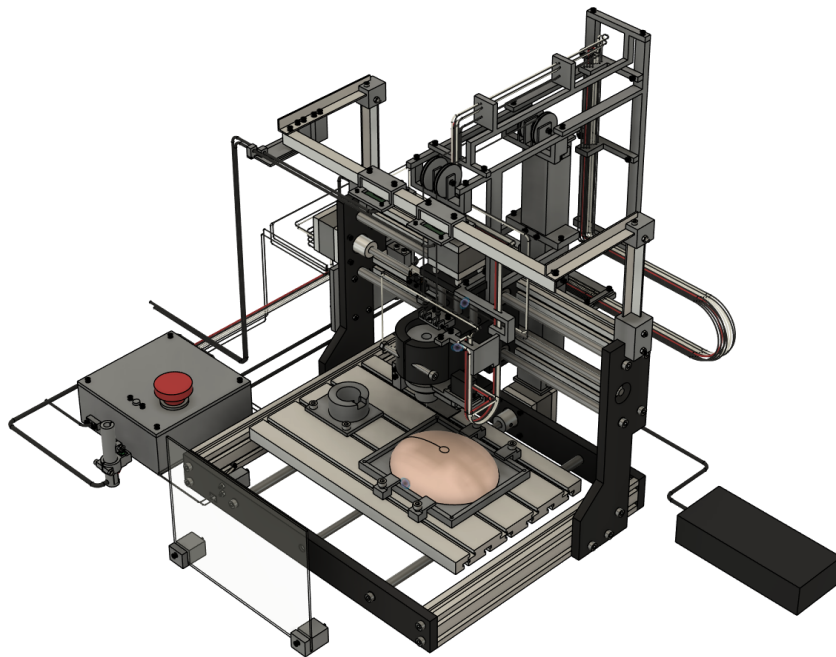


Acuivect

Precision Insertion Demonstrator

Harrison B. Goldberg · September 2026

Digital design complete; physical build deferred due to budget.



Actual native CAD view. This image shows the digital assembly, not a physical prototype.

A benchtop insertion demonstrator combining XYZ positioning, camera-to-tool alignment and a removable cartridge on a synthetic brain phantom.

Independent portfolio work for a Neuralink application. No company affiliation or endorsement.
Synthetic-phantom bench demonstrator only.

AI-assisted workflows supported the creation of this project.

System design

The second revision integrates custom tooling and instrumentation around a commercial motion-platform kit. The private native model retains editable geometry and its engineering records.

System	Nominal design reference
Motion platform	Genmitsu 3018-PRO kit with custom cartridge, dock and mounting geometry
Nominal modeled travel	X: 250 mm total; Y: 140 mm total; Z: 45 mm downward
Nominal phantom	120 x 100 x 26 mm; synthetic test medium
Insertion-depth targets	2, 5 and 8 mm; intended test depths, not measured results
Targeting architecture	Two USB cameras; camera-to-tool calibration and physical targeting validation pending
Tool handling	Removable seated cartridge, servo latch and table-mounted dock
Digital assembly	723 occurrences; 395 unique component definitions
Parametric definition	21,616 editable parameters retained in the private native model
Mechanism graph	28 joints, 19 rigid groups and 8 motion links
Print-oriented design	69 unique printed shapes, 84 occurrences; physical prints pending

Design decisions

Cost and reuse. The kit supplies the frame and XYZ drives. Custom parts adapt it to the seated cartridge, dock, cameras and electronics.

Coordinate consistency. The dock and phantom share the moving Y table. Target registration must account for that moving reference frame.

Service and motion. Exploded service views communicate assembly. Digital joint and docking checks separately address operating motion.

Digital evidence and limits

Review	Recorded result	Qualification limit
Native geometry and materials	Passed recorded digital checks	Source-limited purchased geometry still needs incoming measurement.
Motion and contact review	30 sampled critical poses passed	Sampling does not prove clearance at every possible pose or under deformation.
Parameter edit and restore	Passed a representative actual edit/restore	Later interface changes require renewed checks of the affected assembly.
Cartridge docking	Dock, withdrawal and reseating checks passed in CAD	Real latching, friction, retention and sensor behavior are untested.
Print geometry	69 current shapes passed digital geometry/slicing diagnostics	No physical printing, fit or process qualification completed.
Head linear-static study	Reviewed native solve completed	Single current mesh, ideal boundary conditions and isotropic PETG reference.
Physical prototype	Deferred due to budget	No measured targeting accuracy, repeatability, insertion result or hardware stability claim.

Head structural reference

The ideal fixed-clamp reference applies 5 N in Z, 1 N in X and -35 N mm about Y. With an isotropic PETG model, the reviewed result is **0.279 MPa** peak stress, **0.0077216 mm** maximum displacement and **71.662** minimum safety factor.

The revised geometry has one 0.75 mm parabolic-mesh solve. No current peak-stress convergence claim is made. The result does not establish whole-machine accuracy or actual print strength. The thin probe is not qualified by the head load case.

Printed anisotropy, clamp grip, backlash, friction, wiring, stop behavior, physical targeting and long-term stability require hardware testing. Counterbalance is not a brake.

Budget and next stage

The physical build is deferred because the required budget is not yet available. The digital design and review documentation are complete; physical construction and testing have not begun.

Historical budget category	USD
Motion-platform kit	\$159.00
Two cameras	\$59.98
PETG stock, 2 kg	\$27.98
Remaining controls, hardware and consumables	\$242.14
Parts total	\$489.10
Optional delivery reserve	\$50.00
Planning total with reserve	\$539.10

Prices recorded September 9, 2026. Tax, tariffs and shipping are excluded from the parts total. The \$400 goal was exceeded by \$89.10. Existing computer, printer and workshop tools are assumed. This is not a current quotation.

When funding is available

Measure purchased parts and calibrate printed fits; assemble and commission the mechanism and electronics; then measure cartridge retention, stop behavior, camera calibration, positioning repeatability and synthetic-phantom insertion.

Public review documents

General assembly (2 sheets), assembly and motion sequence (2 sheets), and head structural review (3 pages) accompany this portfolio. The source engineering set remains private, including fabrication drawings, editable CAD and manufacturing files.

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